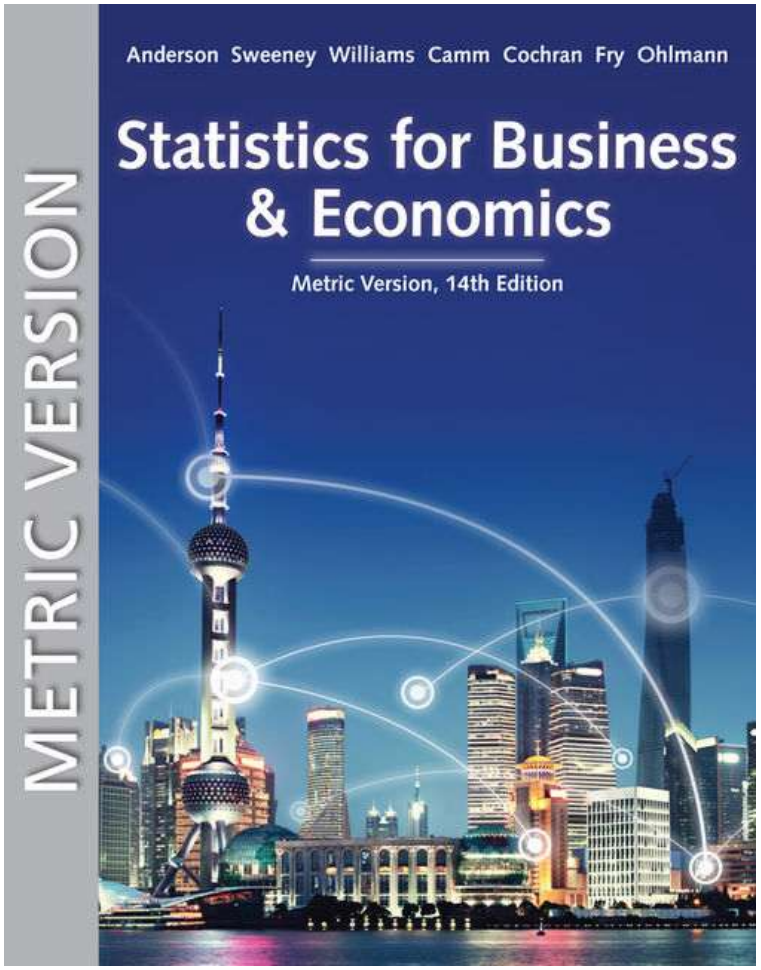
Statistics for Business and Economics (14e) Metric Version

Anderson, Sweeney, Williams, Camm, Cochran, Fry, Ohlmann

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Chapter 2 - Descriptive Statistics: Tabular and Graphical Displays



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Chapter 2 - Descriptive Statistics: Tabular and Graphical Displays

2.1 - Summarizing Data for a Categorical Variable

- Categorical data use labels or names to identify categories of like items.

2.2 - Summarizing Data for a Quantitative Variable

- Quantitative data are numerical values that indicate how much or how many.

Categorical data → Which group or category?
Quantitative data → How much or how many?

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2.3 - Summarizing Data for Two Variables Using Tables

2.4 - Summarizing Data for Two Variables Using Graphical Displays

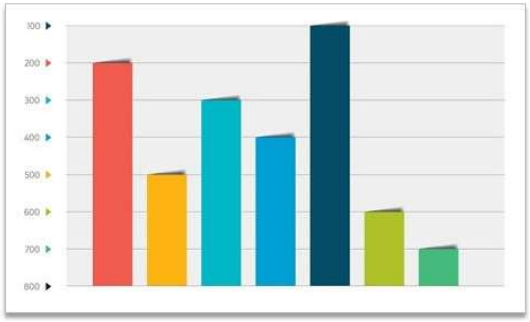
2.5 - Data Visualization: Best Practices in Creating Effective Graphical Displays

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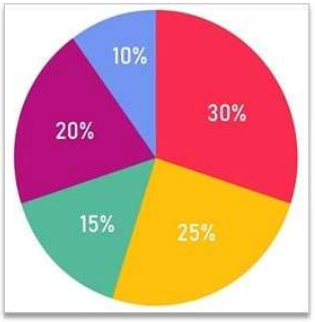
Summarizing Categorical Data

- Frequency Distribution
- Relative Frequency Distribution
- Percent Frequency Distribution
- Bar Chart
- Pie Chart

Grades	Tally	Frequency
66-75	𝍸	5
76-85	𝍸	5
86-95	𝍸 IIII	9
96-105	I	1



VS.



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Frequency Distribution

A frequency distribution is a **tabular** summary of data showing the number (frequency) of observations in each of several **non-overlapping** categories or classes.

Example: Marada Inn

Guests staying at the Marada Inn were asked to rate the quality of their accommodations as being *excellent*, *above average*, *average*, *below average*, or *poor*.

Rating	Frequency
Poor	2
Below Average	3
Average	5
Above Average	9
Excellent	1
Total	20

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Which hotel received more “Excellent” ratings?



Hotel A

👥 20 responses

⭐ 10 “Excellent”

10 / 20 = 50%



Hotel B

👥 200 responses

⭐ 50 “Excellent”

50 / 200 = 25%

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Relative Frequency and Percent Frequency Distributions

- The relative frequency of a class is the fraction or proportion of the total number of data items belonging to the class.

$$\text{Relative frequency} = \frac{\text{Frequency}}{n}$$

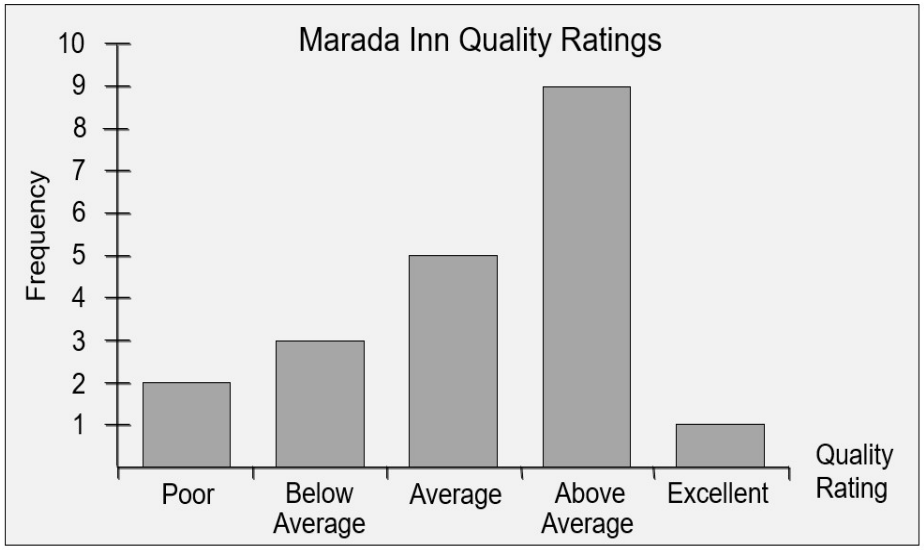
- The percent frequency of a class is the relative frequency multiplied by 100.

Rating	Frequency	Relative Frequency	Percent Frequency
Poor	2	0.10	10%
Below Average	3	0.15	15%
Average	5	0.25	25%
Above Average	9	0.45	45%
Excellent	1	0.05	5%
Total	20	1.00	100%

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Bar Chart

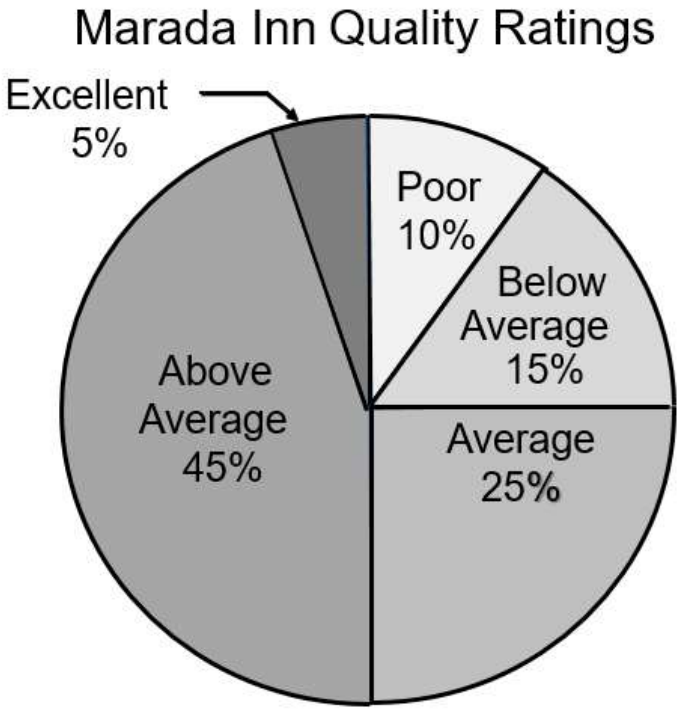
- A bar chart is a graphical display for depicting qualitative data.
- A frequency, relative frequency, or percent frequency scale can be used for the other axis (usually the vertical axis).
- Using a bar of fixed width drawn above each class label, we extend the height appropriately.
- The bars are separated to emphasize the fact that each class is a separate category.



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Pie Chart

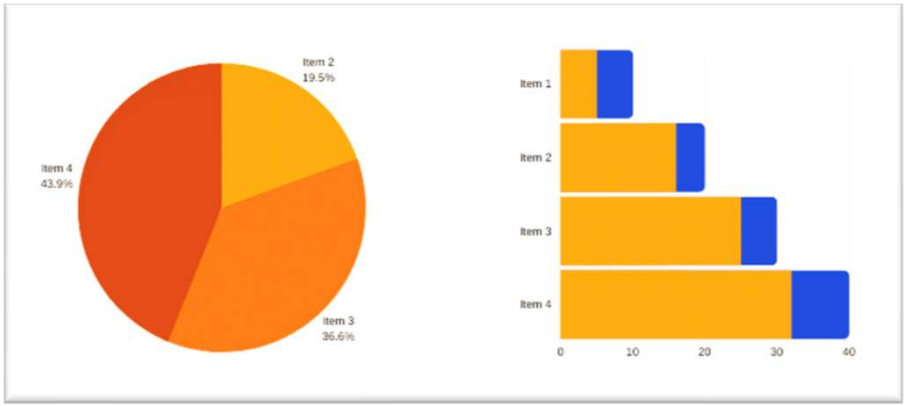
- The pie chart is a commonly used graphical display for presenting relative frequency and percent frequency distributions for categorical data.
- First draw a circle; then use the relative frequencies to subdivide the circle into sectors that correspond to the relative frequency for each class.
- Because there are 360 degrees in a circle, a class with a relative frequency of 0.25 would consume 0.25 x 360 = 90 degrees of the circle.

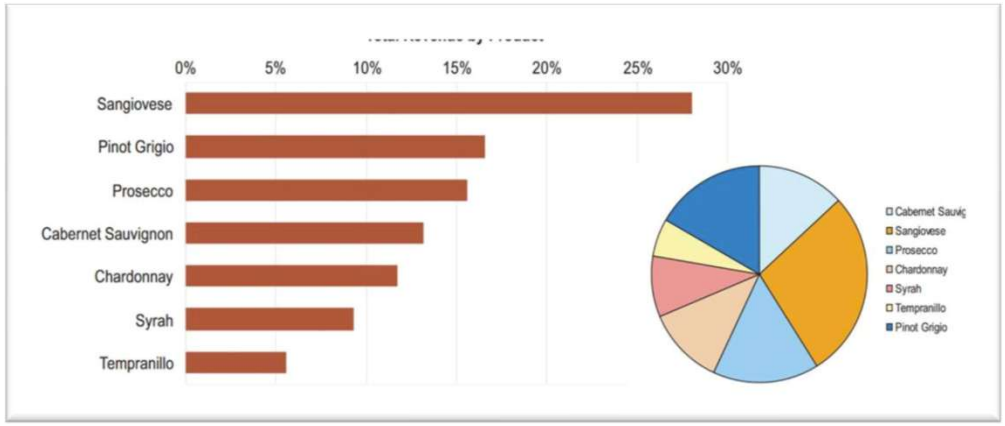


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When Should We Use a Pie Chart?

✅ Works Well	⚠️ Be Careful
Few categories	Too many categories
Clear differences	Similar percentages
Shows **parts of a whole**	Slices are hard to compare





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Summarizing Quantitative Data

- Frequency Distribution
- Relative Frequency and Percent Frequency Distributions
- Dot Plot
- Histogram
- Cumulative Distributions
- Stem-and-Leaf Display

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Frequency Distribution – Quantitative Data

Example

The manager of Hudson Auto would like to gain a better understanding of the cost of parts used in the engine tune-ups performed in the shop. She examines 50 customer invoices for tune-ups. The costs of parts, rounded to the nearest dollar, are shown below.

Sample of Parts Cost($) for 50 Tune-ups

91	78	93	57	75	52	99	80	97	62
71	69	72	89	66	75	79	75	72	76
104	74	62	68	97	105	77	65	80	109
85	97	88	68	83	68	71	69	67	74
62	82	98	101	79	105	79	69	62	73

- For categorical data, we count observations in each category.
- For quantitative data, we count observations in each class or range.

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Frequency Distribution – Quantitative Data

Example: Hudson Auto Repair

If we choose six classes the approximate class width = (109 – 52)/6 = 9.5 or about 10.

Sample of Parts Cost($) for 50 Tune-ups

91	78	93	57	75	52	99	80	97	62
71	69	72	89	66	75	79	75	72	76
104	74	62	68	97	105	77	65	80	109
85	97	88	68	83	68	71	69	67	74
62	82	98	101	79	105	79	69	62	73

Part Cost ($)	Frequency
50-59	2
60-69	13
70-79	16
80-89	7
90-99	7
100-109	5
Total	50

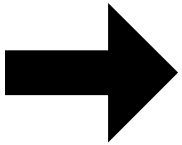
Class Width ≈ (Highest Value – Lowest Value) ÷ Number of Classes

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Relative Frequency and Percent Frequency Distributions

- Relative Frequency = Frequency ÷ Total Number of Items
- Percent Frequency = Relative Frequency × 100

Part Cost ($)	Frequency
50-59	2
60-69	13
70-79	16
80-89	7
90-99	7
100-109	5
Total	50



Parts Cost ($)	Relative Frequency	Percent Frequency
50-59	0.04 = 2/50	4 = .04(100)
60-69	0.26	26
70-79	0.32	32
80-89	0.14	14
90-99	0.14	14
100-109	0.10	10
Total	1.00	100

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Insights

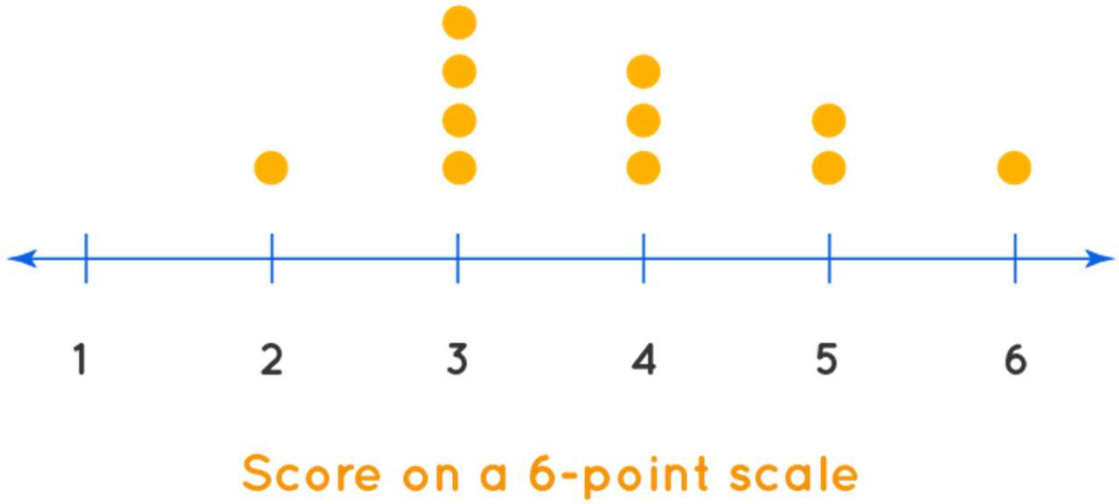
- Only 4% of the parts costs are in the $50-59 class.
- 30% of the parts costs are under $70.
- The greatest percentage (32% or almost one-third) of the parts costs are in the $70-79 class.
- 10% of the parts costs are $100 or more.

Parts Cost ($)	Relative Frequency	Percent Frequency
50-59	0.04 = 2/50	4 = .04(100)
60-69	0.26	26
70-79	0.32	32
80-89	0.14	14
90-99	0.14	14
100-109	0.10	10
Total	1.00	100

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Dot Plot

- One of the simplest graphical summaries of data is a dot plot.
- A horizontal axis shows the range of data values.
- Then each data value is represented by a dot placed above the axis.

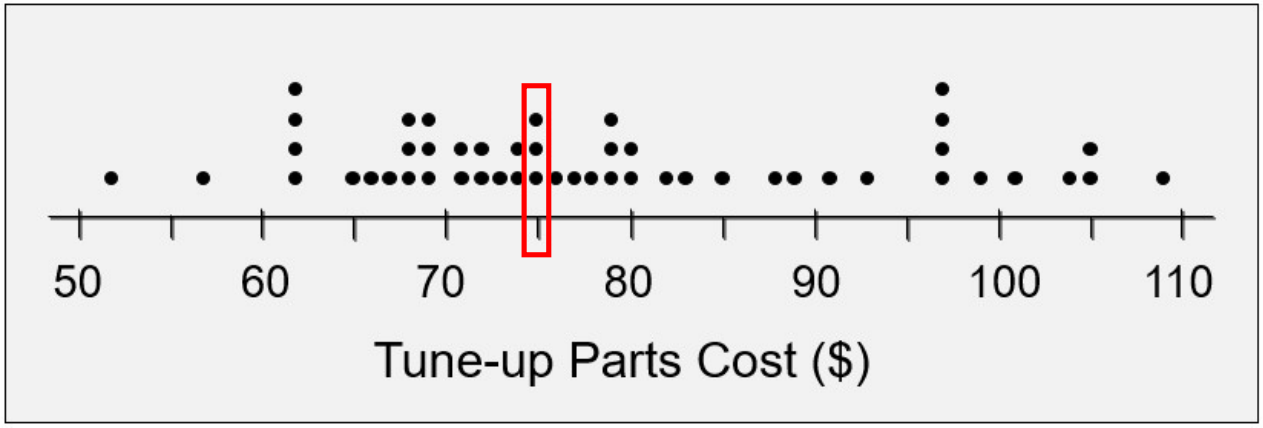


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Dot Plot

Sample of Parts Cost($) for 50 Tune-ups

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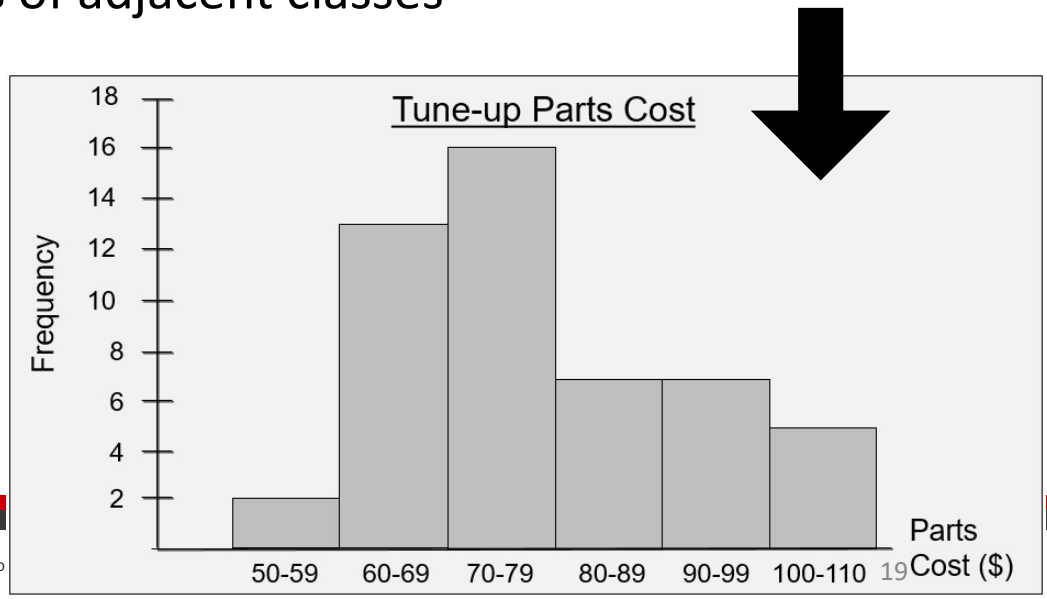


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Histogram (直方圖)

- The variable of interest is placed on the horizontal axis
- A rectangle is drawn above each class interval
- The height of the rectangle represents the interval's frequency (or relative frequency/percent frequency)
- No natural separation between rectangles of adjacent classes

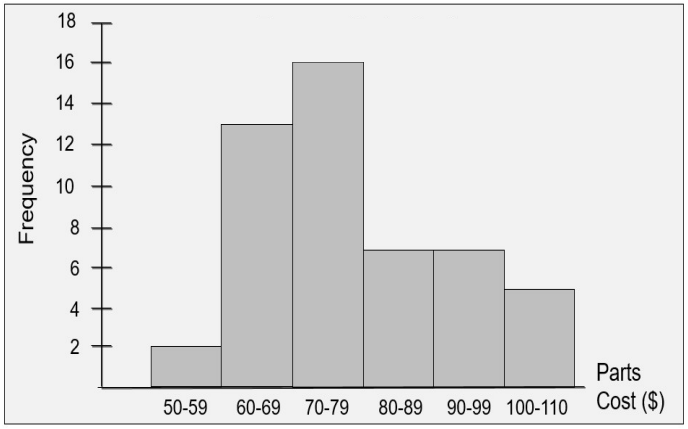
Part Cost ($)	Frequency
50-59	2
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80-89	7
90-99	7
100-109	5
Total	50

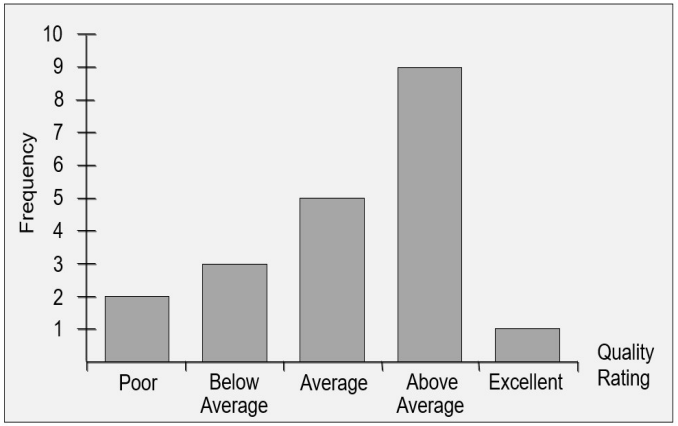


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Differences between a histogram and a bar chart

	Histogram	**Bar chart**
Type of data	Quantitative (numerical) data	Categorical (qualitative) data
Bar appearance	Bars touch each other	Bars are separated



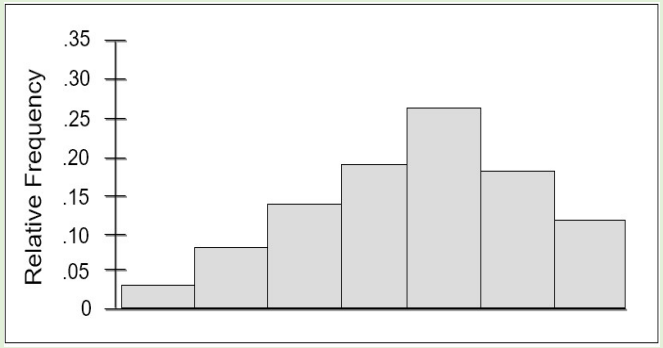


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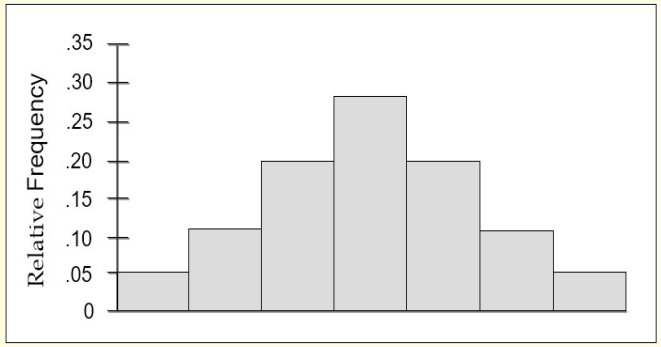
Histograms Showing Skewness (偏度)

- Skewness tells us whether a distribution is balanced or pulled to one side.

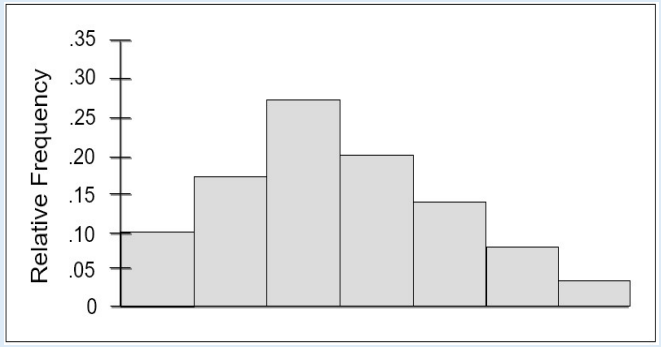
Left Skewed (左偏態)



Symmetric (對稱)



Right Skewed (右偏態)

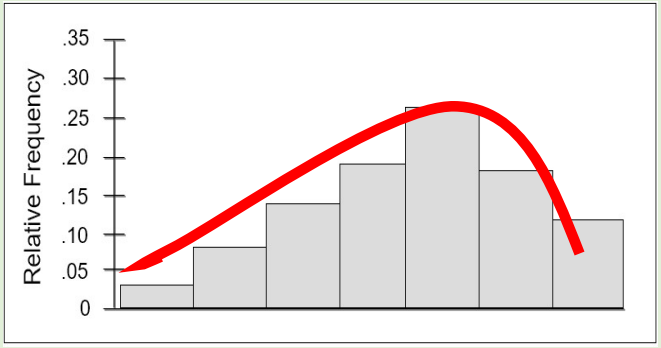


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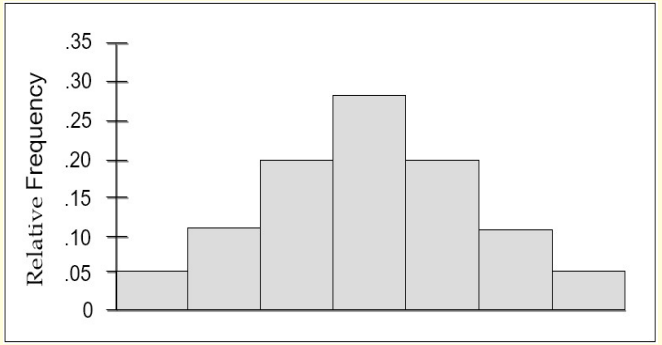
Histograms Showing Skewness (偏度)

Left Skewed (左偏態)

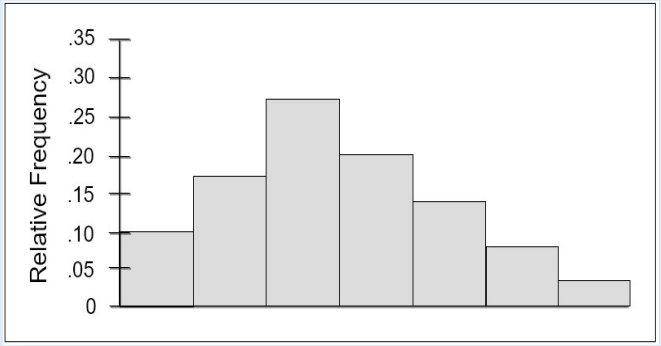
- **Skewness < 0**
  (Negative skewness)
- A longer tail to the left
- Ex: Scores on an easy exam



Symmetric (對稱)



Right Skewed (右偏態)

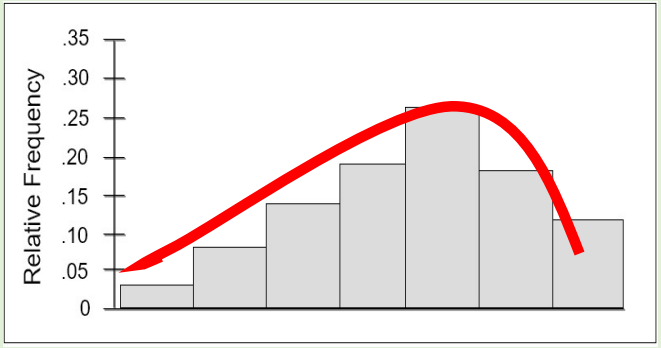


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Histograms Showing Skewness (偏度)

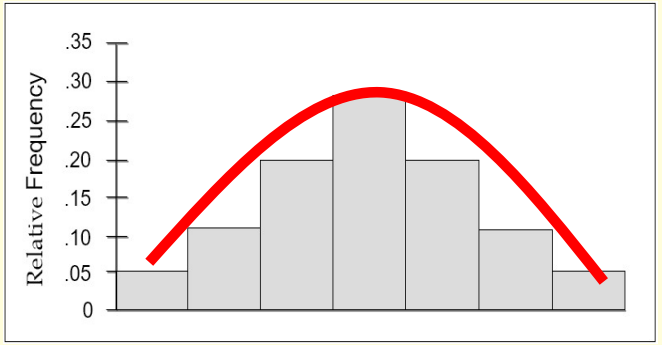
Left Skewed (左偏態)

- **Skewness < 0**
  (Negative skewness)
- A longer tail to the left
- Ex: Scores on an easy exam

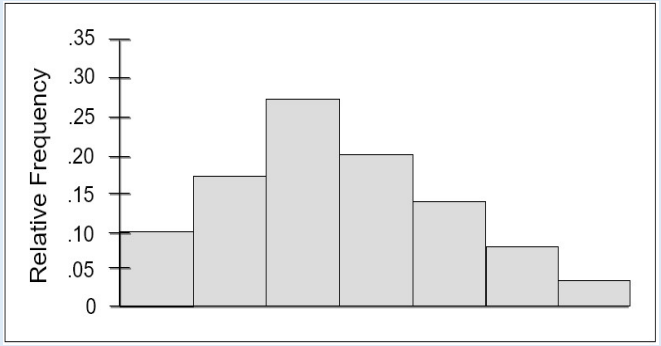


Symmetric (對稱)

- **Skewness = 0**
- Left tail is the mirror image of the right tail
- Ex: Heights of People



Right Skewed (右偏態)

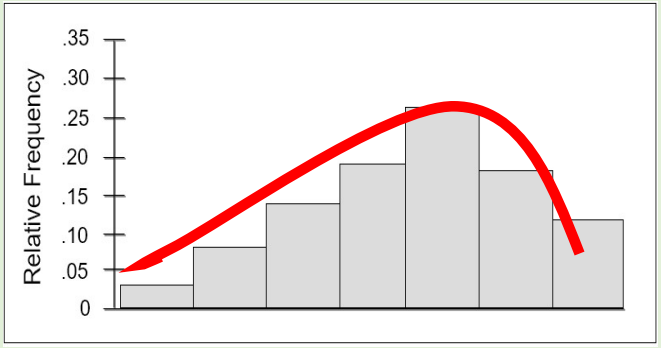


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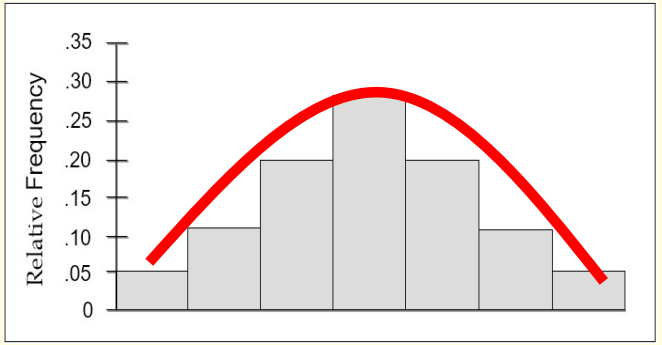
Left Skewed (左偏態)

- **Skewness < 0**
  (Negative skewness)
- A longer tail to the left
- Ex: Scores on an easy exam



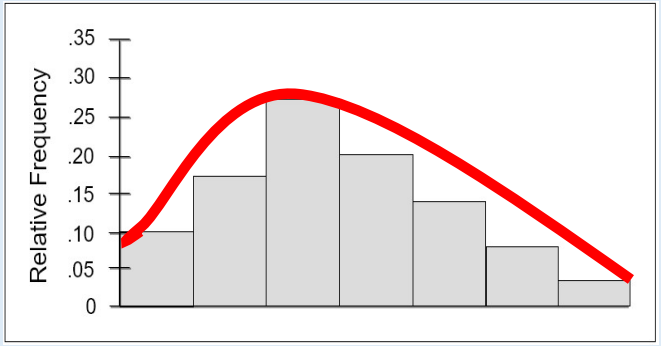
Symmetric (對稱)

- **Skewness = 0**
- Left tail is the mirror image of the right tail
- Ex: Heights of People



Right Skewed (右偏態)

- **Skewness > 0**
  (Positive skewness)
- A longer tail to the right
- Ex: Online game spending



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Cumulative Distributions

Cumulative frequency distribution – shows the *number* of items with values less than or equal to the upper limit of each class.

Cumulative relative frequency distribution – shows the *proportion* of items with values less than or equal to the upper limit of each class.

Cumulative percent frequency distribution – shows the *percentage* of items with values less than or equal to the upper limit of each class.

Hudson Auto Repair

Part Cost ($)	Frequency
50-59	2
60-69	13
70-79	16
80-89	7
90-99	7
100-109	5
Total	50

Cost ($)	Cumulative Frequency	Cumulative Relative Frequency	Cumulative Percent Frequency
≤ 59	2		
≤ 69	15 = 2+13		
≤ 79	31		
≤ 89	38		
≤ 99	45		
≤ 109	50		

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Hudson Auto Repair

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70-79	16
80-89	7
90-99	7
100-109	5
Total	50

Cost ($)	Cumulative Frequency	Cumulative Relative Frequency	Cumulative Percent Frequency
≤ 59	2	.04	4
≤ 69	15 = 2+13	.30 = 15/50	30 = .30(100)
≤ 79	31	.62	62
≤ 89	38	.76	76
≤ 99	45	.90	90
≤ 109	50	1.00	100

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Cumulative Distributions

Part Cost ($)	Frequency
50-59	2
60-69	13
70-79	16
80-89	7
90-99	7
100-109	5
Total	50

- Cumulative Frequency → How many so far?
- Cumulative Relative Frequency → What proportion so far?
- Cumulative Percent Frequency → What percentage so far?

Cost ($)	Cumulative Frequency	Cumulative Relative Frequency	Cumulative Percent Frequency
≤ 59	2	.04	4
≤ 69	15 = 2+13	.30 = 15/50	30 = .30(100)
≤ 79	31	.62	62
≤ 89	38	.76	76
≤ 99	45	.90	90
≤ 109	50	1.00	100

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Stem-and-Leaf Display (莖葉圖)

- A stem-and-leaf display shows both the rank order and shape of a distribution of data.
- It is similar to a histogram on its side, but it has the advantage of showing the actual data values.
- The leading digits of each data item are arranged to the left of a vertical line.
- To the right of the vertical line we record the last digit for each item in rank order.

Sample of Parts Cost($) for 50 Tune-ups

91	78	93	57	75	52	99	80	97	62
71	69	72	89	66	75	79	75	72	76
104	74	62	68	97	105	77	65	80	109
85	97	88	68	83	68	71	69	67	74
62	82	98	101	79	105	79	69	62	73

Stems	Leaves
5	2 7
6	2 2 2 2 5 6 7 8 8 8 9 9 9
7	1 1 2 2 3 4 4 5 5 5 6 7 8 9 9 9
8	0 0 2 3 5 8 9
9	1 3 7 7 7 8 9
10	1 4 5 5 9

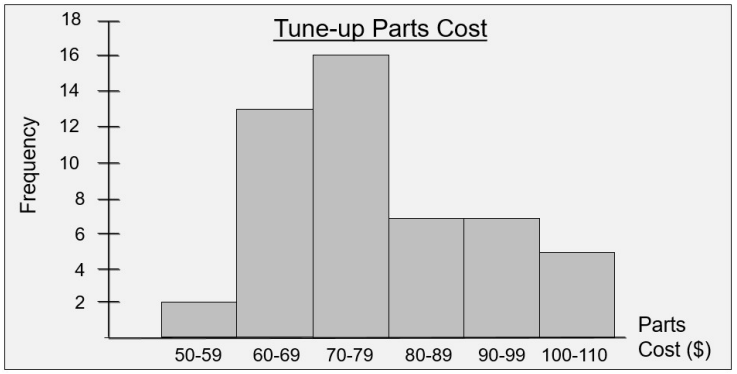
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Stem-and-Leaf Display (莖葉圖)

- Each line (row) in the display is referred to as a stem.
- Each digit lined up on a stem is a leaf.

Stems	Leaves
5	2 7
6	2 2 2 2 5 6 7 8 8 8 9 9 9
7	1 1 2 2 3 4 4 5 5 5 6 7 8 9 9 9
8	0 0 2 3 5 8 9
9	1 3 7 7 7 8 9
10	1 4 5 5 9

Shows the shape of the distribution
&
Keeps the original data values



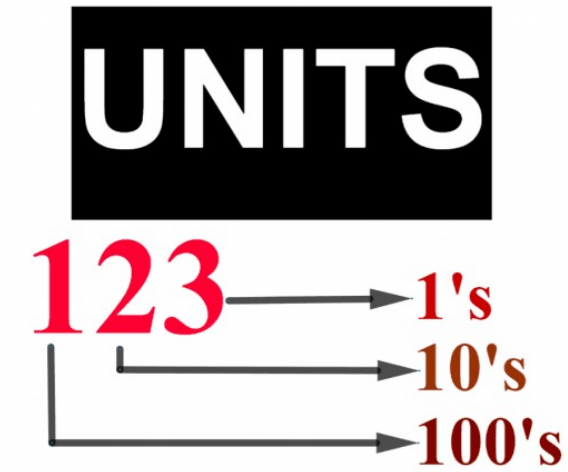
Shows the shape of the distribution

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Stem-and-Leaf Display (莖葉圖)

Leaf Units

- A single digit is used to define each leaf.
- In the preceding example, the leaf unit was 1.
- Leaf units may be 100, 10, 1, 0.1, and so on.
- Where the leaf unit is not shown, it is assumed to equal 1.
- The leaf unit indicates how to multiply the stem-and-leaf numbers in order to approximate the original data.



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Stem-and-Leaf Display (莖葉圖)

Leaf Units

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- Leaf units may be 100, 10, 1, 0.1, and so on.
- Where the leaf unit is not shown, it is assumed to equal 1.
- The leaf unit indicates how to multiply the stem-and-leaf numbers in order to approximate the original data.

If we have data with values such as
8.6 11.7 9.4 9.1 10.2 11.0 8.8

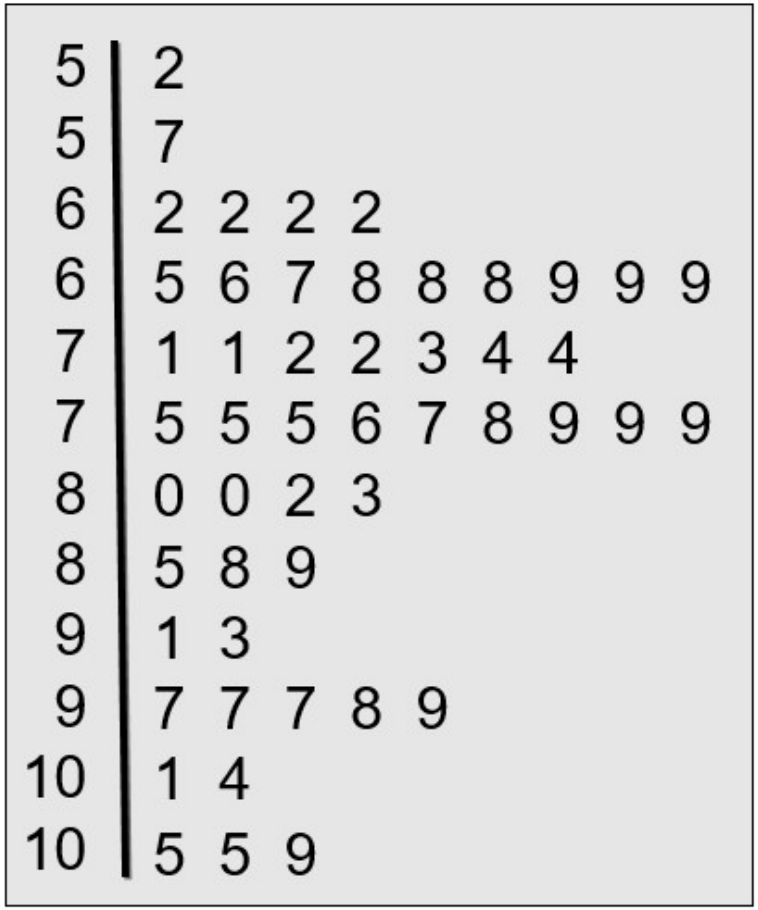
Leaf Unit = 0.1

8	6 8
9	1 4
10	2
11	0 7

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Stretched Stem-and-Leaf Display

- If we believe the original stem-and-leaf display has condensed the data too much, we can stretch the display vertically by using two stems for each leading digit(s).
- Whenever a stem value is stated twice, the first value corresponds to leaf values of 0 - 4, and the second value corresponds to leaf values of 5 - 9.



5	2
5	7
6	2 2 2 2
6	5 6 7 8 8 8 9 9 9
7	1 1 2 2 3 4 4
7	5 5 5 6 7 8 9 9 9
8	0 0 2 3
8	5 8 9
9	1 3
9	7 7 7 8 9
10	1 4
10	5 5 9

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Stem-and-Leaf Display

If we have data values such as
91, 93, 94, 95, 97, and 98

Regular Stem-and-leaf Display

Stem	Leaf
9	1 3 4 5 7 8

Stretched Stem-and-Leaf Display

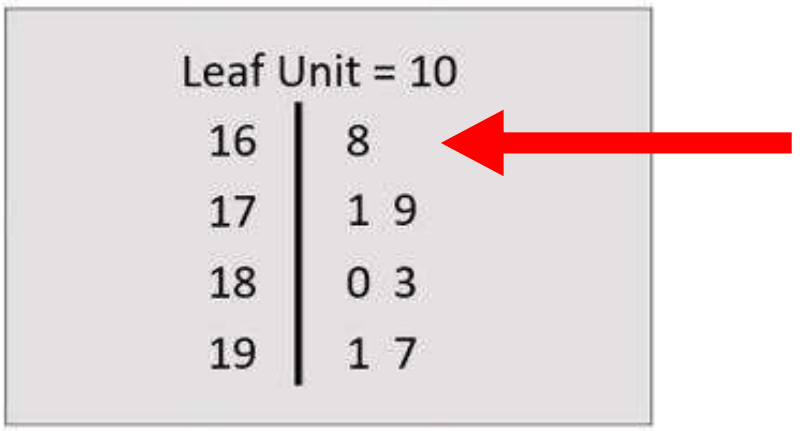
Stem	Leaf
9	1 3 4
9	5 7 8

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Stem-and-Leaf Display

If we have data values such as

1806, 1717, 1974, 1791, 1682, 1910, and 1838



The 82 in 1682 is rounded down to 80 and is represented as an 8.

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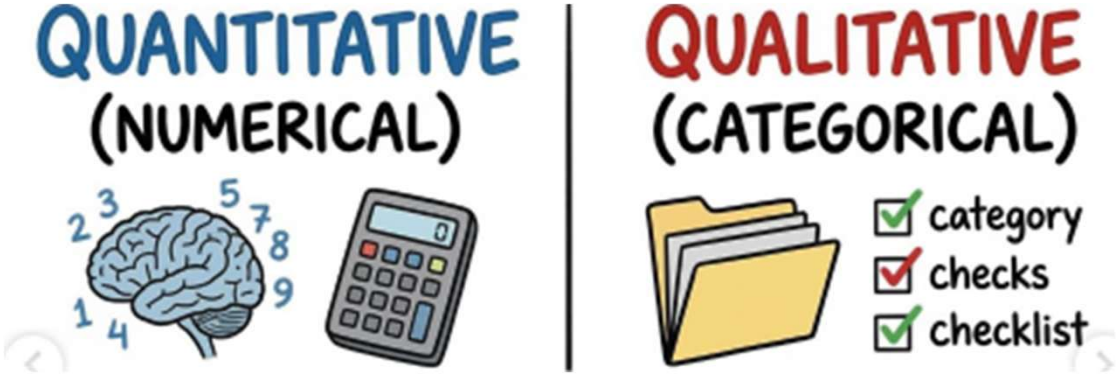
Summarizing Data for Two Variables using Tables

- Thus far we have focused on methods that are used to summarize the data for one variable at a time.
- Often a manager is interested in tabular and graphical methods that will help understand the relationship between two variables.
- Crosstabulation (交叉表格) is a method for summarizing the data for two variables.
- The left and top margin labels define the classes for the two variables.

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Summarizing Data for Two Variables using Tables

- Crosstabulation can be used when:
  - one variable is categorical and the other is quantitative,
  - both variables are categorical, or
  - both variables are quantitative



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Crosstabulation

Example: Finger Lakes Homes

The number of Finger Lakes homes sold for each style and price for the past two years is shown below.

Price Range	Home Style				Total
	Colonial	Log	Split	A-Frame	
< $250,000	18	6	19	12	55
≥ $250,000	12	14	16	3	45
Total	30	20	35	15	100

Insights

- The greatest number of homes (19) in the sample are a split-level style and priced at less than $250,000.
- Only three homes in the sample are an A-Frame style and priced at $250,000 or more.

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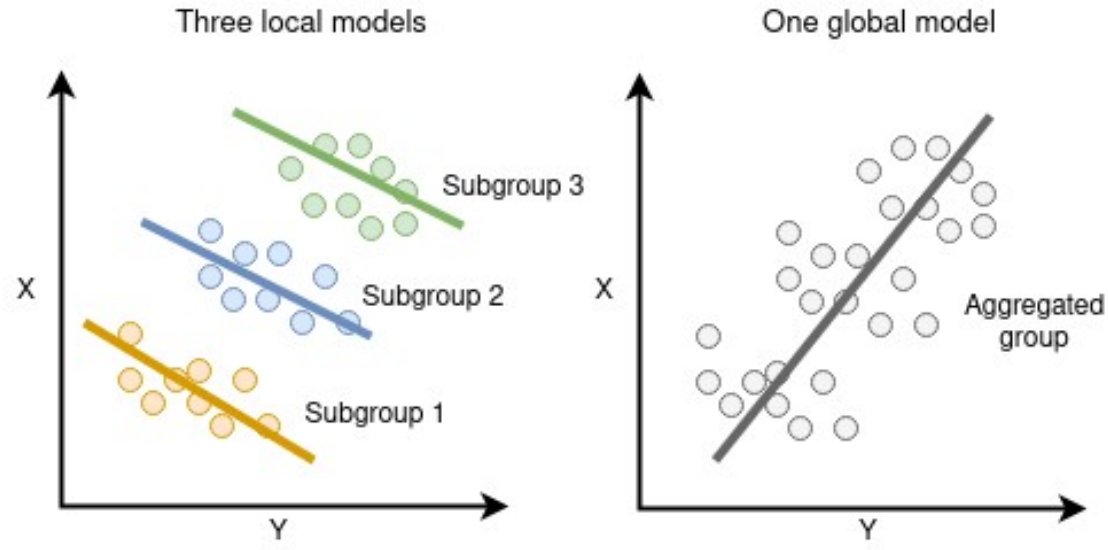
Crosstabulation: Simpson's Paradox

- Data in two or more crosstabulations are often aggregated to produce a summary crosstabulation.
- We must be careful in drawing conclusions about the relationship between the two variables in the aggregated crosstabulation.
- In some cases the conclusions based upon an aggregated crosstabulation can be completely reversed if we look at the unaggregated data. The reversal of conclusions based on aggregate and unaggregated data is called Simpson's paradox.

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Crosstabulation: Simpson's Paradox

- Imagine a company evaluating the productivity of two employees, Alan and John, based on the number of tasks they complete in two projects - Project 1 and Project 2.



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Crosstabulation: Simpson's Paradox

- Imagine a company evaluating the productivity of two employees, Alan and John, based on the number of tasks they complete in two projects - Project 1 and Project 2

	All Tasks (Project 1)	Completed Tasks (Project 1)	Completion Rate (Project 1)	All Tasks (Project 2)	Completed Tasks (Project 2)	Completion Rate (Project 2)	Combined Completion Rate	Normalized Completion Rate
John	48	12	25.0%	582	183	31.4%	31.0%	28.2%
Jane	411	104	25.3%	140	45	32.1%	27.0%	28.7%

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Summarizing Data for Two Variables Using Graphical Displays

- In most cases, a graphical display is more useful than a table for recognizing patterns and trends.
- Displaying data in creative ways can lead to powerful insights.
- Scatter diagrams (散佈圖) and trendlines are useful in exploring the relationship between two variables.

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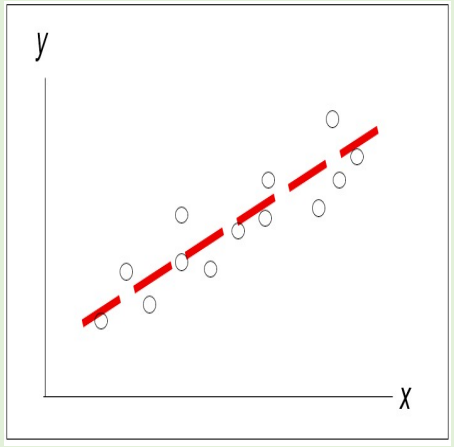
Scatter Diagram and Trendline

- A scatter diagram is a graphical presentation of the relationship between two quantitative variables.
- One variable is shown on the horizontal axis and the other variable is shown on the vertical axis.
- The general pattern of the plotted points suggests the overall relationship between the variables.
- A trendline provides an approximation of the relationship.

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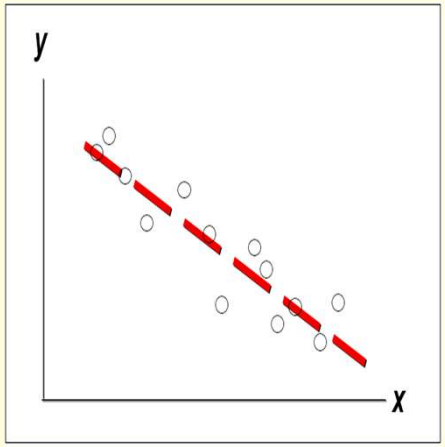
Scatter Diagram

A Positive Relationship

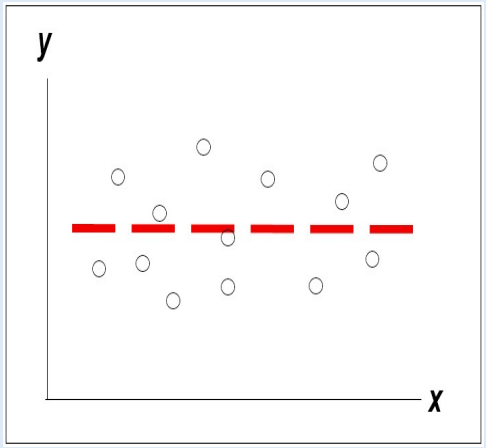


- Both variables tend to increase together.
- Ex: Study hours and exam scores

A negative Relationship



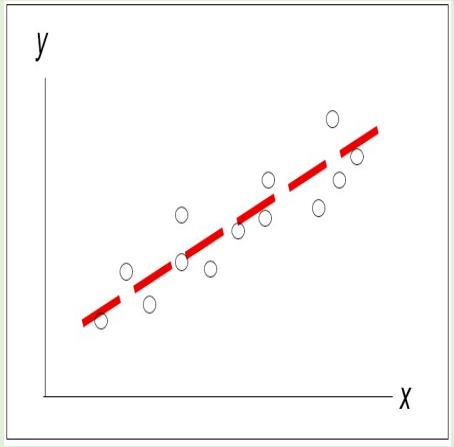
No Apparent Relationship



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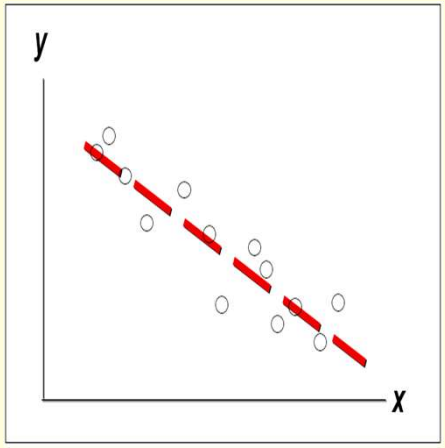
Scatter Diagram

A Positive Relationship



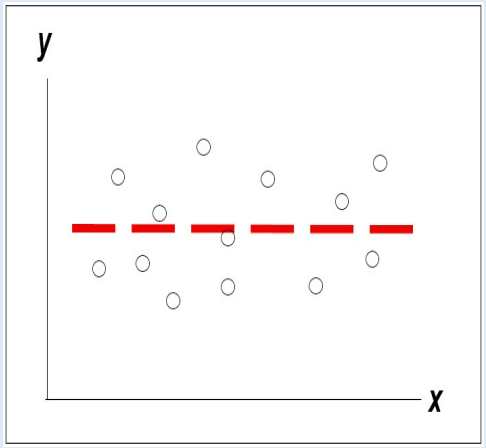
- Both variables tend to increase together
- Ex: Study hours & exam scores

A negative Relationship



- One tends to increase as the other decreases
- Ex: Price of a product & the quantity demanded

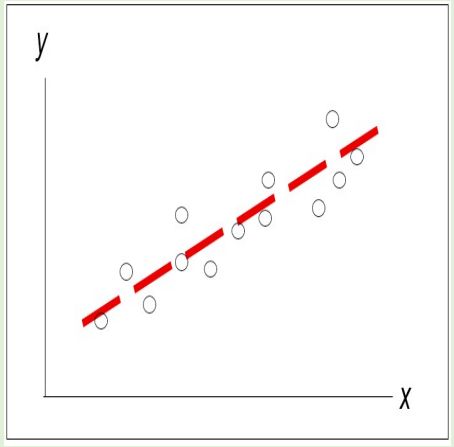
No Apparent Relationship



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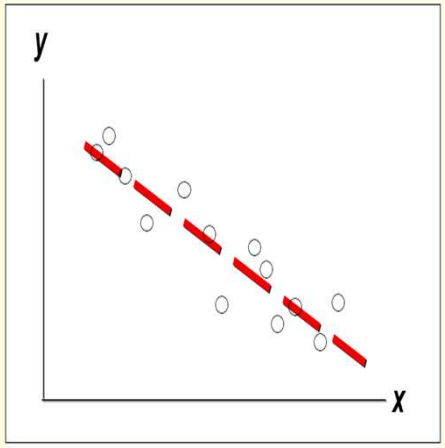
Scatter Diagram

A Positive Relationship



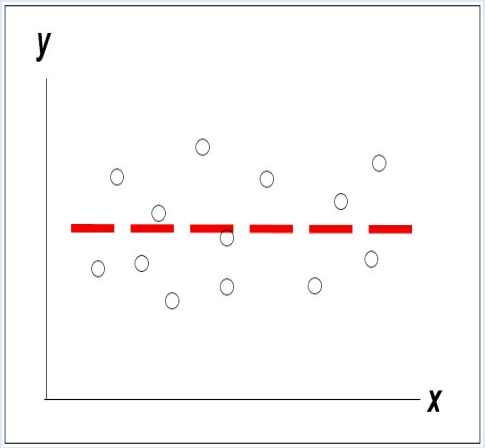
- Both variables tend to increase together
- Ex: Study hours & exam scores

A negative Relationship



- One tends to increase as the other decreases
- Ex: Price of a product & the quantity demanded

No Apparent Relationship



- No clear pattern between the variables
- Ex: Shoe size & exam scores

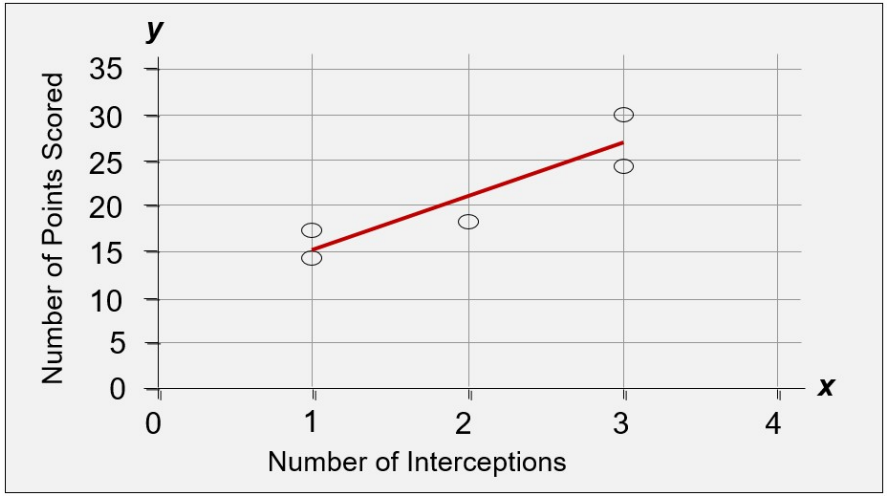
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Scatter Diagram

Example: Panthers Football Team

The Panthers football team is interested in investigating the relationship, if any, between interceptions made and points scored.

X = Number of Interceptions	Y = Number of Points Scored
1	14
3	24
2	18
1	17
3	30



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Example: Panthers Football Team

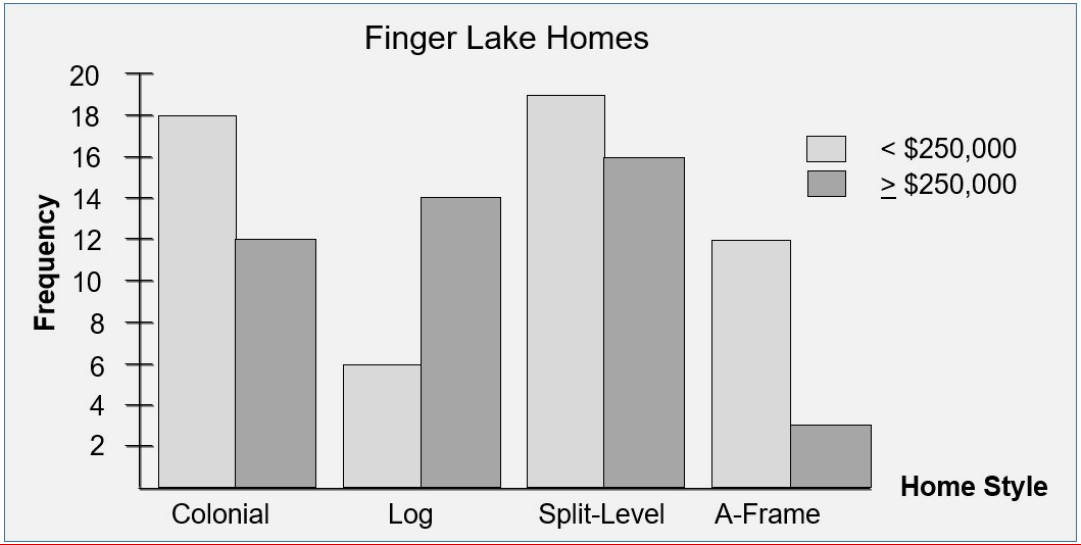
Insights gained from the preceding Scatter Diagram

- The scatter diagram indicates a positive relationship between the number of interceptions and the number of points scored.
- Higher points scored are associated with a higher number of interceptions.
- The relationship is not perfect; all plotted points in the scatter diagram are not on a straight line.

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Side-by-Side Bar Chart (群組長條圖)

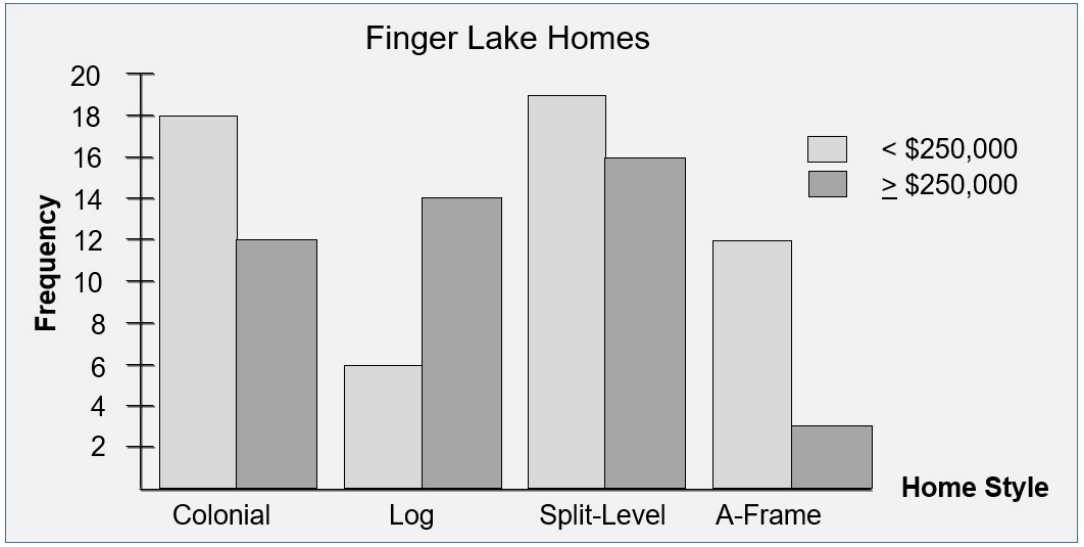
- A side-by-side bar chart is a graphical display for depicting multiple bar charts on the same display.
- Each cluster of bars represents one value of the first variable.
- Each bar within a cluster represents one value of the second variable.



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Side-by-Side Bar Chart (群組長條圖)

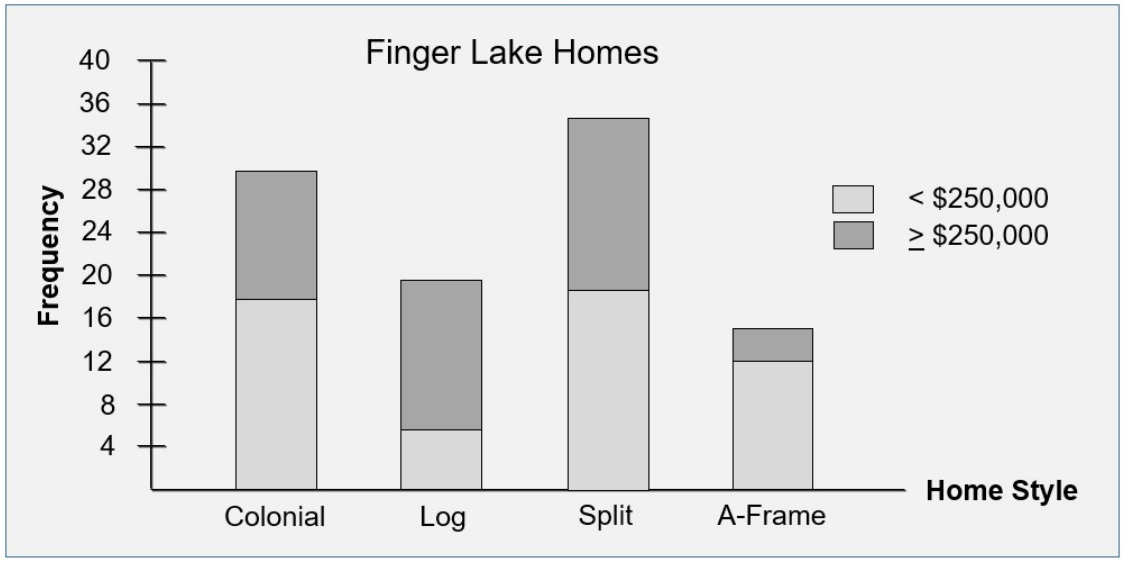
- Each cluster represents a particular home style
- Two bars
  - One for homes priced under $250,000
  - One for homes priced $250,000 or more



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Stacked Bar Chart (堆疊長條圖)

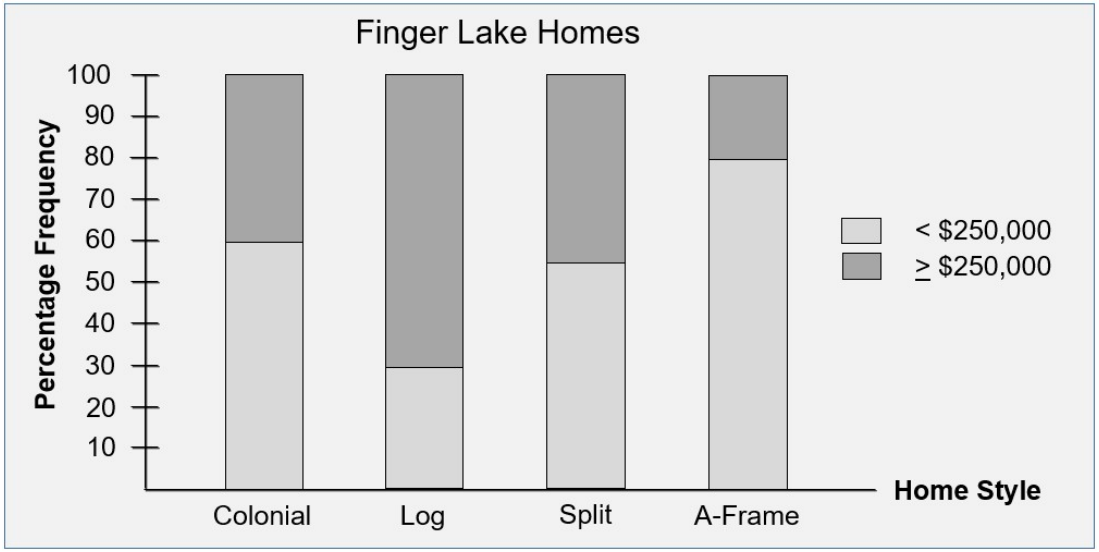
- A stacked bar chart is another way to display and compare two variables on the same display.
- It is a bar chart in which each bar is broken into rectangular segments of a different color.



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Stacked Bar Chart (堆疊長條圖)

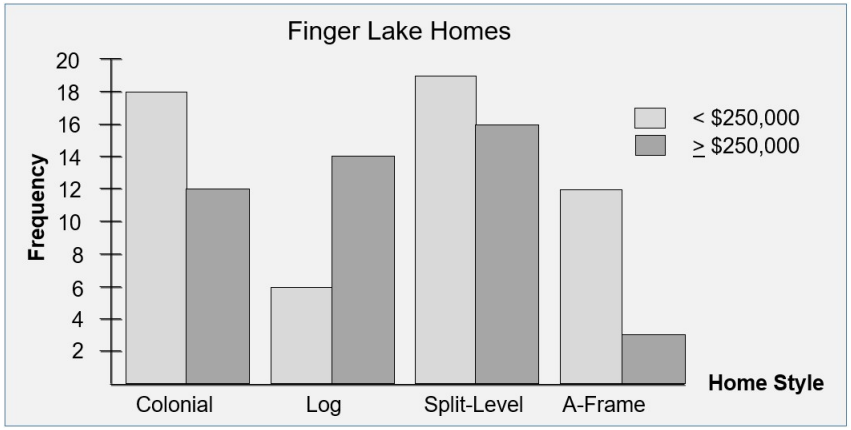
If percentage frequencies are displayed, all bars will be of the same height (or length), extending to the 100% mark.

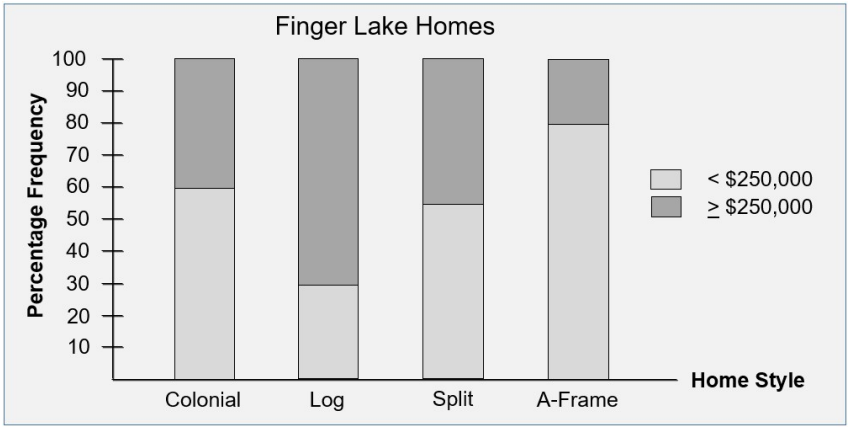


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Differences between a side-by-side bar chart and a stacked bar chart

Side-by-Side Bar Chart	Stacked Bar Chart
Compare individual groups	See the total
Differences are easy to compare	See the composition of the total





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Data Visualization:
Best Practices in Creating Effective Graphical Displays

- Data visualization is the use of graphical displays to summarize and present information about a data set.
- The goal is to communicate, as effectively and clearly as possible, the key information about the data.

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Choosing the Type of Graphical Display

Displays used to show the distribution of data:

- Bar Chart to show the frequency distribution and relative frequency distribution for categorical data
- Pie Chart to show the relative frequency and percent frequency for categorical data
- Dot Plot to show the distribution for quantitative data over the entire range of the data
- Histogram to show the frequency distribution for quantitative data over a set of class intervals
- Stem-and-Leaf Display to show both the rank order and shape of the distribution for quantitative data

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Choosing the Type of Graphical Display

Displays used to make comparisons:

- Side-by-Side Bar Chart to compare two variables
- Stacked Bar Chart to compare the relative frequency or percent frequency of two categorical variables

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Choosing the Type of Graphical Display

Displays used to show relationships:

- Scatter Diagram to show the relationship between two quantitative variables
- Trendline to approximate the relationship of data in a scatter diagram

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Data Dashboards

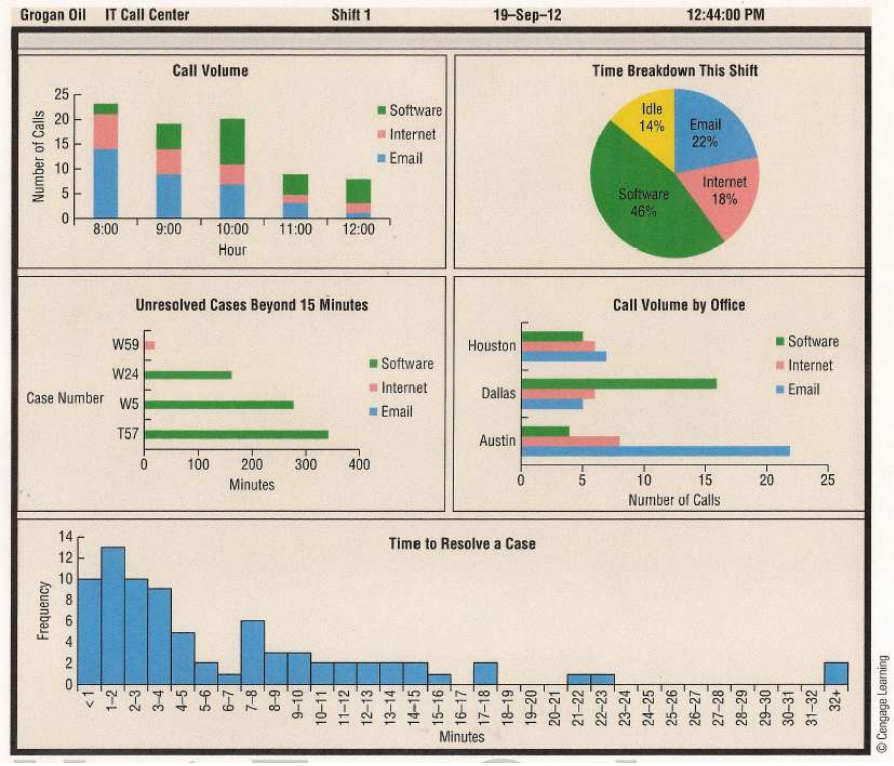
- A data dashboard is a widely used data visualization tool.
- It organizes and presents key performance indicators (KPIs) used to monitor an organization or process.
- It provides timely summary information that is easy to read, understand, and interpret.



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Data Dashboards

- Some additional guidelines include:
  - Minimize the need for screen scrolling.
  - Avoid unnecessary use of color or 3D displays.
  - Use borders between charts to improve readability.



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